



# Advanced Time Series Analysis and Forecasting

## Seminar 6: State space models and Bayesian filtering



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## Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **APF:** Auxiliary Particle Filter
- ▣ **AR:** AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARIMA:** AutoRegressive Integrated Moving Average (model)
- ▣ **ARMA:** AutoRegressive Moving Average (model)
- ▣ **BN:** Beveridge–Nelson (decomposition)
- ▣ **BNR:** Banca Națională a României (the National Bank of Romania)
- ▣ **CEE:** Central and Eastern Europe
- ▣ **DFM:** Dynamic Factor Model
- ▣ **DK:** Durbin–Koopman (simulation smoother, 2002)
- ▣ **ESS:** Effective Sample Size
- ▣ **FFBS:** Forward Filtering, Backward Sampling
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP:** Gross Domestic Product
- ▣ **HICP:** Harmonised Index of Consumer Prices

## Acronyms used in this material (2/2)

- ▣ **IG**: inverse gamma (distribution)
- ▣ **IMA**: Integrated Moving Average (model)
- ▣ **KSC**: Kim–Shephard–Chib (1998), mixture sampler for stochastic volatility
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MA**: Moving Average
- ▣ **ML**: Machine Learning
- ▣ **MNZ**: Morley–Nelson–Zivot (2003)
- ▣ **PMMH**: Particle Marginal Metropolis–Hastings
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **SCA**: Seasonally and Calendar Adjusted
- ▣ **SV**: Stochastic Volatility
- ▣ **TSA**: Time Series Analysis
- ▣ **TVP**: Time-Varying Parameter(s)
- ▣ **UC**: Unobserved Components (model)
- ▣ **UC-SV**: Unobserved Components model with Stochastic Volatility (Stock–Watson 2007)
- ▣ **US**: United States
- ▣ **VAT**: Value Added Tax

## Today's question and route

- **Question:** how do we estimate what we cannot observe (a level, a trend, a volatility, a gap), and how do we tell a real signal from the noise of the estimator?
  - ▶ this seminar comes **before** Lecture 6: the section “Prerequisites for Today” gives every definition the tasks use
- Route
  - ▶ Part A: the exact diffuse filter by hand, the concentrated likelihood, backward sampling, the Gibbs block, SV moments, one particle-filter step, the APF, Beveridge–Nelson
  - ▶ Part B: Romanian trend growth; TVP regressions for Hungary and Poland; SV of EUR/RON by Gibbs and by the particle filter; the real-time Romanian output gap
  - ▶ Part C: is Romanian trend inflation anchored? An AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

## Exercise map

| Task    | Question                                                              | Type             | Model      |
|---------|-----------------------------------------------------------------------|------------------|------------|
| A1, A2  | the exact diffuse Kalman filter by hand; the concentrated likelihood  | Solved, Proposed | A1         |
| A3, A4  | one backward-sampling step; the inverse-gamma Gibbs block             | Solved, Proposed | A3         |
| A5, A6  | moments of the SV model; the QML measurement equation                 | Solved, Proposed | A5         |
| A7, A8  | one bootstrap particle-filter step; auxiliary weights with an outlier | Solved, Proposed | A7         |
| A9, A10 | the Beveridge–Nelson cycle; local level, IMA(1,1) and BN              | Solved, Proposed | A9         |
| B1, B2  | Romanian trend growth; TVP regressions for Hungary and Poland         | Solved, Proposed | B1         |
| B3, B4  | SV of EUR/RON: Gibbs, then the particle filter                        | Solved, Proposed | B3         |
| B5      | the Romanian output gap in real time                                  | Proposed         | B1, A10    |
| C1, C2  | is Romanian trend inflation anchored?; what is wrong in an AI answer? | Proposed         | B3, A1–A10 |

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

## Data used

| Series                                    | Source                 | Frequency | Period    |
|-------------------------------------------|------------------------|-----------|-----------|
| Romania: real GDP (SCA)                   | Eurostat namq_10_gdp   | quarterly | 2000–2026 |
| HICP: Romania, Hungary, Poland, euro area | Eurostat prc_hicp_minr | monthly   | 2001–2026 |
| EUR/RON reference rate                    | BNR                    | daily     | 2016–2026 |

- In the notebook: `kalman_filter`, `kalman_smoother`, `sim_smoother_dk`, `ffbs`, `sv_gibbs`, `pf_sv`, `fit_llt`, `fit_tvp`, `fit_smooth_trend`, `ucsv_gibbs`; no account or key is needed

## Prerequisites for Today (1/9): the state space model

- Model (TSA, Chapter 10): observations are a noisy linear function of a hidden state that follows a linear dynamic

$$y_t = Z_t \alpha_t + \varepsilon_t, \varepsilon_t \sim N(0, H); \quad \alpha_{t+1} = T \alpha_t + R \eta_t, \eta_t \sim N(0, Q)$$

- Notation

- ▶  $y_t$ : the observation;  $\alpha_t$ : the unobserved state;  $Z_t$ : measurement matrix;  $T$ : transition matrix;  $R$ : selects the states with shocks
- ▶  $H$ : variance of the measurement noise  $\varepsilon_t$ ;  $Q$ : covariance of the state shocks  $\eta_t$ ;  $n$ : the sample length
- ▶  $Y_t = (y_1, \dots, y_t)$ ;  $a_t = \mathbb{E}(\alpha_t | Y_{t-1})$ ,  $P_t = \text{Var}(\alpha_t | Y_{t-1})$ : the predicted state and its variance

- Local level example:  $Z = T = R = 1$ ,  $y_t = \mu_t + \varepsilon_t$ ,  $\mu_{t+1} = \mu_t + \eta_t$

## Prerequisites for Today (2/9): the Kalman filter

- Prediction error and its variance:  $v_t = y_t - Z_t a_t$ ,  $F_t = Z_t P_t Z_t' + H$ 
  - ▶  $v_t$ : the new information in  $y_t$ ; it is independent of the past data
- Kalman gain  $K_t = TP_t Z_t' / F_t$ : how much of the surprise  $v_t$  moves the next state prediction
  - ▶  $a_{t+1} = Ta_t + K_t v_t$ ,  $P_{t+1} = TP_t(T - K_t Z_t)' + RQR'$
  - ▶ filtered state (after seeing  $y_t$ ):  $a_{t|t} = a_t + P_t Z_t' v_t / F_t$
- Log-likelihood:  $\sum_t -\frac{1}{2}(\ln 2\pi + \ln F_t + v_t^2 / F_t)$ , a by-product of the filter

## Prerequisites for Today (3/9): the exact diffuse start

- Nonstationary states (random walks) have no stationary variance:  $P_1 = \kappa P_\infty + P_*$ ,  $\kappa \rightarrow \infty$ 
  - ▶  $P_\infty$ : marks the diffuse elements;  $P_*$ : covariance of the others;  $F_{\infty,t} = Z_t P_{\infty,t} Z_t'$  and  $F_{*,t}$ : the diffuse and finite parts of  $F_t$
- While  $F_{\infty,t} > 0$ , use two gains:  $K^{(0)} = TP_\infty Z' / F_\infty$ ,  $K^{(1)} = T(P_* Z' - P_\infty Z' F_* / F_\infty) / F_\infty$ 
  - ▶  $a_{t+1} = Ta_t + K^{(0)} v_t$ ,  $P_{\infty,t+1} = TP_\infty L^{(0)'}$ ,  $P_{*,t+1} = TP_\infty L^{(1)'} + TP_* L^{(0)'} + RQR'$
  - ▶  $L^{(0)} = T - K^{(0)}Z$ ,  $L^{(1)} = -K^{(1)}Z$ ; after  $d$  diffuse steps  $P_\infty = 0$  and the ordinary filter takes over
- Diffuse log-likelihood: the diffuse steps contribute  $-\frac{1}{2}(\ln 2\pi + \ln F_{\infty,t})$ , the others  $-\frac{1}{2}(\ln 2\pi + \ln F_t + v_t^2 / F_t)$ 
  - ▶ scale concentrated out: if  $H, Q$  are proportional to  $\sigma^2$ , then  $\hat{\sigma}^2 = \frac{1}{n-d} \sum_{t>d} v_t^2 / F_t^*$ ,  $F_t^* = F_t / \sigma^2$

## Prerequisites for Today (4/9): sampling the states

- ▣ FFBS (Carter–Kohn): run the filter forward, then draw the states backward
  - ▶ draw  $\alpha_n \sim N(a_{n|n}, P_{n|n})$ ; then  $\alpha_t | \alpha_{t+1} \sim N(a_{t|t} + G_t(\alpha_{t+1} - Ta_{t|t}), P_{t|t} - G_t TP_{t|t})$
  - ▶  $G_t = P_{t|t} T'(TP_{t|t} T' + RQR')^{-1}$ : how much the next state tells us about the current one
- ▣ Durbin–Koopman: simulate  $(\alpha^+, y^+)$  from the model, smooth  $y - y^+$  once:  $\tilde{\alpha} = \alpha^+ + \hat{\alpha}(y - y^+)$ 
  - ▶  $\hat{\alpha}(\cdot)$ : the smoothed state mean;  $\tilde{\alpha}$ : an exact draw of the whole state path

## Prerequisites for Today (5/9): Gibbs blocks and the boundary

- Gibbs block for a variance with prior  $IG(a_0, b_0)$  and  $m$  shocks  $e_t$ :  $\sigma^2 | e \sim IG(a_0 + m/2, b_0 + \sum e_t^2/2)$ 
  - ▶  $IG(a, b)$ : inverse gamma with shape  $a$  and scale  $b$ , mean  $b/(a - 1)$ ;  $e_t$ : the shocks implied by the current state draw
- Inefficiency factor  $1 + 2 \sum_k \rho_k$ : how many correlated draws are worth one independent draw
  - ▶  $\rho_k$ : autocorrelation of the chain at lag  $k$
- Pile-up: the ML estimate of a variance can be exactly zero even when the true value is positive; LR tests at the boundary need a bootstrap

## Prerequisites for Today (6/9): stochastic volatility

- SV:  $y_t = e^{h_t/2}\epsilon_t$ ,  $h_t = \mu + \phi(h_{t-1} - \mu) + \sigma_\eta\eta_t$ ,  $\epsilon_t, \eta_t \sim N(0, 1)$ 
  - ▶  $h_t$ : log-variance (latent AR(1));  $\mu$ : its mean;  $\phi$ : persistence;  $\sigma_\eta$ : volatility of volatility
  - ▶  $\sigma_h^2 = \sigma_\eta^2/(1 - \phi^2)$ : the variance of  $h_t$ ; kurtosis  $3e^{\sigma_h^2} > 3$
  - ▶  $\text{Corr}(y_t^2, y_{t-k}^2) = (e^{\sigma_h^2\phi^k} - 1)/(3e^{\sigma_h^2} - 1)$ : volatility clustering
- Linear form:  $\ln y_t^2 = h_t + \ln \epsilon_t^2$ , with  $\mathbb{E} \ln \epsilon^2 = -1.2704$ ,  $\text{Var} \ln \epsilon^2 = \pi^2/2$ 
  - ▶ KSC: a seven-normal mixture for  $\ln \epsilon^2$ ; given the mixture indicators,  $h$  is Gaussian and the Kalman tools apply

## Prerequisites for Today (7/9): particle filters

- Bootstrap particle filter:  $N$  particles  $x_t^{(i)}$  approximate the distribution of the state
  - ▶ propagate  $x_t^{(i)} \sim f(\cdot | x_{t-1}^{(i)})$  ( $f$ : transition density); weight  $w_t^{(i)} = g(y_t | x_t^{(i)})$  ( $g$ : measurement density)
  - ▶ likelihood term  $\hat{p}(y_t | Y_{t-1}) = \frac{1}{N} \sum w_t^{(i)}$ ; normalised weights  $W_t^{(i)} = w_t^{(i)} / \sum_j w_t^{(j)}$ ; ESS =  $1 / \sum (W_t^{(i)})^2$ ; then resample
- Systematic resampling:  $u_i = (U + i - 1) / N$ , one uniform  $U$ 
  - ▶ particle  $j$  is copied for every  $u_i$  in  $(C_{j-1}, C_j]$ ,  $C_j$  = cumulative normalised weights
- Auxiliary filter (Pitt–Shephard): first-stage weights  $\propto g(y_t | \mu_t^{(i)})$ ,  $\mu_t^{(i)} = \mathbb{E}(x_t | x_{t-1}^{(i)})$ ; then propagate and correct

## Prerequisites for Today (8/9): trend and cycle

- Beveridge–Nelson trend:  $\tau_t = y_t + \sum_{j \geq 1} \mathbb{E}_t(\Delta y_{t+j} - \mu)$ 
  - ▶ the level where  $y$  settles once the predictable growth dies out;  $\mu$ : the mean growth
  - ▶ AR(1) in growth:  $\Delta y_t - \mu = \phi(\Delta y_{t-1} - \mu) + e_t$
  - ▶ then the cycle is  $c_t = y_t - \tau_t = -\frac{\phi}{1-\phi}(\Delta y_t - \mu)$
- Local level with signal-to-noise ratio  $q = \sigma_\eta^2 / \sigma_\varepsilon^2$ 
  - ▶  $\Delta y_t$  is MA(1):  $\Delta y_t = u_t + \theta u_{t-1}$ ,  $u_t$  white noise
  - ▶  $\theta = -(1 + q/2) + \sqrt{q + q^2/4}$
- Local linear trend:  $\mu_{t+1} = \mu_t + \nu_t + \xi_t$ ,  $\nu_{t+1} = \nu_t + \zeta_t$ 
  - ▶  $\nu_t$ : the slope;  $\xi_t$ ,  $\zeta_t$ : the level and slope shocks
  - ▶  $4\nu_t$ : the annualised trend growth of a quarterly log series

## Prerequisites for Today (9/9): output gaps and trend inflation

- Smooth-trend UC:  $y_t = \tau_t + c_t$ ,  $\Delta^2 \tau_{t+1} = \zeta_t$ ,  $c_t$  AR(2)
  - ▶ filtered gap  $c_{t|t}$  (real time) against smoothed gap  $\hat{c}_t$  (full sample)
- UC-SV (Stock–Watson):  $\pi_t = \tau_t + \sigma_{\eta,t}\eta_t$ ,  $\tau_t = \tau_{t-1} + \sigma_{\varepsilon,t}\varepsilon_t$ 
  - ▶ inflation  $\pi_t = \text{trend } \tau_t + \text{transitory noise; } \eta_t, \varepsilon_t \sim N(0, 1)$
  - ▶  $\ln \sigma_{\eta,t}^2$  and  $\ln \sigma_{\varepsilon,t}^2$  follow random walks with variance  $\gamma$

## A1: the exact diffuse filter by hand [Solved]

## Task

- ▣ Local level,  $\sigma_\varepsilon^2 = 1$ ,  $\sigma_\eta^2 = 0.5$ ,  $\mu_1$  diffuse ( $a_1 = 0$ ,  $P_{\infty,1} = 1$ ,  $P_{*,1} = 0$ ); data  $y = (2.0; 3.0; 2.5)$ .
- ▣ 1. Run the diffuse step at  $t = 1$ : give  $F_\infty$ ,  $K^{(0)}$ ,  $K^{(1)}$ ,  $a_2$ ,  $P_{\infty,2}$ ,  $P_{*,2}$ .
- ▣ 2. Run the ordinary filter for  $t = 2, 3$ :  $v_t$ ,  $F_t$ ,  $K_t$ ,  $a_{t+1}$ ,  $P_{t+1}$ .
- ▣ 3. Compute the diffuse log-likelihood and say why  $y_1$  carries no information about the variances.
- ▣ Report: one diffuse step, two ordinary steps, one number, one sentence.

## Solution

- ▣ 1.  $F_\infty = 1$ ,  $F_* = 0 + 1 = 1$ ;  $K^{(0)} = 1$ ,  $K^{(1)} = 0 - 1 \cdot 1/1 = -1$ ;  
 $a_2 = 0 + 1 \cdot 2 = 2.000$ ;  $P_{\infty,2} = 1(1 - 1) = 0$ ;  
 $P_{*,2} = 1 \cdot 1 + 0 + 0.5 = 1.500$
- ▣ 2.  $t = 2$ :  $v = 1.000$ ,  $F = 2.500$ ,  $K = 0.600$ ,  $a_3 = 2.600$ ,  
 $P_3 = 1.5(1 - 0.6) + 0.5 = 1.100$ ;  $t = 3$ :  $v = -0.100$ ,  
 $F = 2.100$ ,  $K = 0.524$ ,  $a_4 = 2.548$ ,  $P_4 = 1.024$
- ▣ 3. Terms  $-0.919$ ,  $-1.577$ ,  $-1.292$ ;  $\ln L_d = -3.788$ . The first term is  $-\frac{1}{2} \ln 2\pi$  whatever the variances:  $y_1$  only fixes the level

## A2: the concentrated likelihood [Proposed]

Solution discussed in class

### Task

- ▣ The data of A1, now with  $\sigma_\eta^2 = 0.5\sigma_\varepsilon^2$  and  $\sigma_\varepsilon^2$  unknown. Model: A1.
- ▣ 1. Run the filter with  $\sigma_\varepsilon^2 = 1$  and write  $F_t = \sigma_\varepsilon^2 F_t^*$ .
- ▣ 2. Derive  $\hat{\sigma}_\varepsilon^2$  by maximising the diffuse log-likelihood and compute it.
- ▣ 3. Compute the concentrated log-likelihood (without the diffuse term) and state how many parameters remain for the numerical search.
- ▣ Report: two ratios, one estimate, one number, one sentence.

## A3: one backward-sampling step [Solved]

### Task

- ▣ Continue A1: after  $y_2$  the filtered moments are  $a_{2|2}$ ,  $P_{2|2}$ ; suppose the draw for the next period is  $\alpha_3 = 3.0$ .
- ▣ 1. Compute  $a_{2|2}$  and  $P_{2|2}$  from A1.
- ▣ 2. Compute  $G_2$ , the conditional mean  $m_2$  and variance  $S_2$  of  $\alpha_2 | \alpha_3, y_1, y_2$ .
- ▣ 3. Explain why  $y_3$  is not needed in this conditional distribution.
- ▣ Report: two moments, three numbers, one sentence.

### Solution

- ▣ 1.  $a_{2|2} = 2 + 0.6 \cdot 1 = 2.600$ ,  $P_{2|2} = 1.5 - 1.5^2/2.5 = 0.600$
- ▣ 2.  $G_2 = P_{2|2}/(P_{2|2} + \sigma_\eta^2) = 0.545$ ;  
 $m_2 = 2.600 + 0.545(3.0 - 2.600) = 2.818$ ;  
 $S_2 = P_{2|2}(1 - G_2) = 0.273$
- ▣ 3. Markov property: given  $\alpha_3$ , the future observations add nothing about  $\alpha_2$ ; their information already entered through the draw of  $\alpha_3$

## A4: the inverse-gamma Gibbs block [Proposed]

Solution discussed in class

### Task

- ▣ A local level Gibbs sampler,  $n = 50$ ; current state draw gives  $\sum_t (y_t - \mu_t)^2 = 40$  and  $\sum_t (\Delta \mu_t)^2 = 6.2$ ; priors  $IG(2.5, 0.25)$  for both variances. Model: A3.
- ▣ 1. Write the two conditional posteriors with their parameters.
- ▣ 2. Compute their means and the implied  $q$ .
- ▣ 3. Explain why this sampler can never return  $\sigma_\eta^2 = 0$ , unlike ML.
- ▣ Report: two distributions, three numbers, one sentence.

## A5: moments of the SV model [Solved]

### Task

- ▣ SV model with  $\phi = 0.98$ ,  $\sigma_\eta = 0.15$ , Gaussian  $\epsilon_t$ .
- ▣ 1. Compute  $\sigma_h^2$  and the kurtosis of  $y_t$ .
- ▣ 2. Compute  $\text{Corr}(y_t^2, y_{t-k}^2)$  for  $k = 1, 10, 50$  and the half-life of a volatility shock.
- ▣ 3. Explain the difference from GARCH(1,1) with the same persistence.
- ▣ Report: two numbers, four numbers, one sentence.

### Solution

- ▣ 1.  $\sigma_h^2 = 0.0225/0.0396 = 0.568$ ; kurtosis  $3e^{0.568} = 5.30$
- ▣ 2. 0.173, 0.138, 0.054: slow decay from a low start; half-life  $\ln 0.5 / \ln 0.98 = 34.3$  days
- ▣ 3. GARCH variance is known one step ahead; in SV it has its own shock, so excess kurtosis arises even with  $\phi$  close to 1 and the likelihood needs an integral

## A6: the QML measurement equation [Proposed]

Solution discussed in class

### Task

- ▣ The SV model of A5, written as  $\ln y_t^2 = h_t + \xi_t$ . Model: A5.
- ▣ 1. Give the mean and variance of  $\xi_t$  and the signal-to-noise ratio  $\sigma_h^2 / \text{Var} \xi_t$ .
- ▣ 2. Compute  $\text{Corr}(\ln y_t^2, \ln y_{t-k}^2)$  for  $k = 1, 10$  and name the ARMA model of  $\ln y_t^2$ .
- ▣ 3. Explain why the Kalman filter on  $\ln y_t^2$  is consistent but inefficient.
- ▣ Report: three numbers, two numbers and a model, one sentence.

## A7: one step of the bootstrap particle filter [Solved]

## Task

- ▣ SV model,  $N = 4$  particles after propagation:  
 $h^{(i)} = (-1; 0; 0.5; 1.5)$ ; observation  $y_t = 2.0$ ; systematic resampling with  $U = 0.3$ .
- ▣ 1. Compute  $w^{(i)} = N(y_t; 0, e^{h^{(i)}})$  and the normalised weights.
- ▣ 2. Compute the ESS and the likelihood increment  $\hat{p}(y_t | Y_{t-1})$ .
- ▣ 3. Resample: which particles survive, and how many times?
- ▣ Report: eight numbers, two numbers, one list.

## Solution

- ▣ 1.  $w = (0.0029; 0.0540; 0.0924; 0.1206)$ ;  
 $W = (0.011; 0.200; 0.342; 0.447)$
- ▣ 2. ESS = 2.80 of 4;  $\hat{p} = \frac{1}{4} \sum w = 0.0675$ ,  $\ln \hat{p} = -2.696$
- ▣ 3.  $u = (0.075; 0.325; 0.575; 0.825)$  against  
 $C = (0.011; 0.211; 0.553; 1)$ : particles 2, 3, 4, 4; the  
 low-volatility particle dies, the high-volatility one is copied twice

## A8: auxiliary weights with an outlier [Proposed]

Solution discussed in class

### Task

- ▣ Local level,  $\sigma_\varepsilon^2 = 0.5$ ,  $\sigma_\eta^2 = 0.25$ ; equally weighted particles  $x_{t-1}^{(i)} = (0; 1; 2; 3)$ ; outlier  $y_t = 4.0$ . Model: A7.
- ▣ 1. Compute the normalised first-stage weights  $\propto N(y_t; x_{t-1}^{(i)}, \sigma_\varepsilon^2)$  and their ESS.
- ▣ 2. Compute the fully adapted weights  $\propto N(y_t; x_{t-1}^{(i)}, \sigma_\varepsilon^2 + \sigma_\eta^2)$  and compare.
- ▣ 3. Explain what the bootstrap filter would do with the same outlier.
- ▣ Report: four weights and an ESS, four weights, one sentence.

## A9: the Beveridge–Nelson cycle of an ARIMA(1,1,0) [Solved]

## Task

- ▣ Quarterly growth:  $\Delta y_t - 0.8 = 0.35(\Delta y_{t-1} - 0.8) + e_t$ ; this quarter  $\Delta y_t = 2.0$ .
- ▣ 1. Derive  $\sum_{j \geq 1} \mathbb{E}_t(\Delta y_{t+j} - \mu)$  for an AR(1).
- ▣ 2. Compute the BN cycle  $c_t$  and the long-run multiplier of  $e_t$  on the trend.
- ▣ 3. Explain the sign of the cycle after a strong quarter.
- ▣ Report: one formula, two numbers, one sentence.

## Solution

- ▣ 1.  $\mathbb{E}_t(\Delta y_{t+j} - \mu) = \phi^j(\Delta y_t - \mu)$ , sum =  $\frac{\phi}{1-\phi}(\Delta y_t - \mu)$
- ▣ 2.  $\frac{\phi}{1-\phi} = 0.538$ ;  $c_t = -0.538 \cdot 1.2 = -0.646$ ; multiplier  $1/(1-\phi) = 1.538$
- ▣ 3. With positive growth persistence, a strong quarter raises expected future growth: the trend jumps by more than output, so output is *below* trend

## A10: local level, IMA(1,1) and Beveridge–Nelson [Proposed]

Solution discussed in class

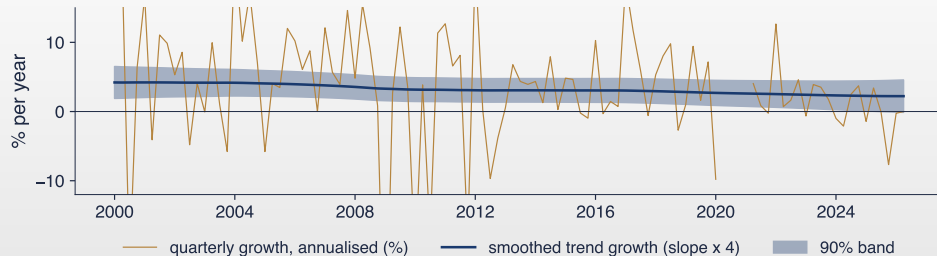
### Task

- ▣ Local level with  $\sigma_\varepsilon^2 = 1$ ,  $q = 0.5$ . Model: A9.
- ▣ 1. Find the MA coefficient  $\theta$  of  $\Delta y_t$ .
- ▣ 2. Solve the steady-state Riccati equation  $P = P - P^2/(P + 1) + q$  and give the gain  $k = P/(P + 1)$ .
- ▣ 3. Show that the BN trend of the IMA(1,1),  $y_t + \theta u_t$ , equals the steady-state filtered level  $a_t|_t$ .
- ▣ Report: one number, two numbers, one derivation.

## B1: Romanian trend growth [Solved]

- ▣ **Question:** has Romania's trend growth fallen since the 2000s, once the uncertainty of the estimate is taken into account?
- ▣ **Inputs:** 100×log real GDP (Eurostat, SCA), 2000Q1–2026Q2,  $T = 106$ ; 2020Q2–Q4 treated as missing
- ▣ **Tasks**
  1. Estimate the local linear trend by exact diffuse ML with the numpy filter and with statsmodels; compare the parameters and the log-likelihoods.
  2. Smooth the slope and report the annualised trend growth  $4\hat{\nu}_t$  with a 90% band in 2005, 2012, 2019 and in the last quarter.
  3. Check whether any variance is at the boundary.
  4. Interpretation: is the fall in trend growth statistically meaningful?
- ▣ **Report:** two parameter sets, four growth rates with bands, one sentence
- ▣ **Notebook:** section B1

## B1: solution [Solved]



- ▣ numpy:  $\sigma_\varepsilon^2 = 0.6013$ ,  $\sigma_\xi^2 = 3.9577$ ,  $\sigma_\zeta^2 = 0.0039$ ,  $\ln L_d = -231.08$ ; statsmodels: 0.6011, 3.9581, 0.0039,  $-229.24$  (the difference in  $\ln L$  is  $\ln 2\pi$ , two diffuse steps)
- ▣ Trend growth: 4.1% (s.d. 1.1) in 2005, 3.1% (1.0) in 2012, 2.8% (1.1) in 2019, 2.2% (1.4) at the end
- ▣ Interpretation: a fall is suggested but not established
  - ▶ the slope variance is small, so trend growth moves slowly
  - ▶ the bands overlap widely; level shocks absorb the 2009 recession

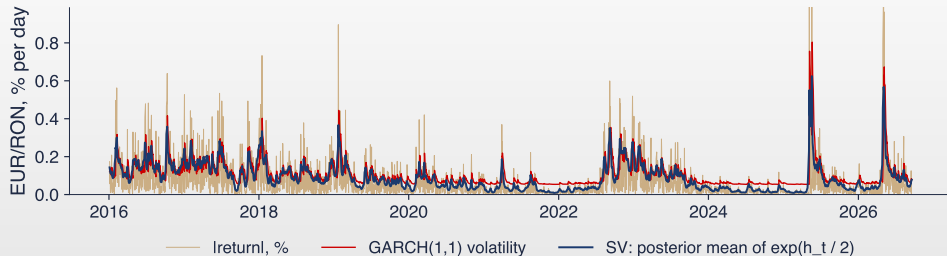
## B2: TVP regressions for Hungary and Poland [Proposed]

- **Question:** has the co-movement of Hungarian and Polish inflation with euro-area inflation changed over time?
- **Inputs:** quarterly HICP inflation (% a.r.), 2005Q4 onward; euro-area inflation net of its seasonal pattern; the model of B1 extended to  $y_t = c_t + b_t x_t + D_t' \gamma + \varepsilon_t$ ; model: B1
- **Tasks**
  1. Estimate the TVP regression by exact diffuse ML (five diffuse states) for each country.
  2. Report  $\hat{\sigma}_b^2$  and the smoothed  $b_t$  with standard errors in 2008Q3, 2019Q4 and 2022Q4.
  3. Compute the LR test of  $\sigma_b^2 = 0$  and its  $\frac{1}{2} \chi_1^2$   $p$ -value; say why this  $p$ -value is only indicative.
  4. Interpretation: is the slope higher than one, and is the co-movement stable or episodic?
- **Report:** a table of two rows, six slopes with standard errors, two tests, one sentence
- **Notebook:** section B2

## B3: stochastic volatility of EUR/RON [Solved]

- **Question:** how persistent is the volatility of the leu against the euro, and how does SV compare with GARCH?
- **Inputs:** daily log returns of the BNR EUR/RON reference rate, %, demeaned, 2016–2026,  $T = 2686$ ; KSC Gibbs sampler with  $c = 10^{-3}\text{Var}(y)$
- **Tasks**
  1. Run the KSC Gibbs sampler and report posterior means and 90% intervals of  $\mu, \phi, \sigma_\eta$  with inefficiency factors.
  2. Estimate a Gaussian GARCH(1,1) on the same data.
  3. Plot the posterior mean of  $\exp(h_t/2)$  against the GARCH volatility.
  4. Interpretation: what does the half-life of volatility say about a managed exchange rate?
- **Report:** a table of three parameters, three GARCH numbers, one chart, one sentence
- **Notebook:** section B3

## B3: solution [Solved]



- ▣  $\phi = 0.957$  [0.944; 0.969],  $\sigma_\eta = 0.541$  [0.488; 0.597],  $\mu = -5.628$  [-6.032; -5.210]; inefficiency factors 18, 42, 1 (20 000 draws)
- ▣ GARCH(1,1):  $\alpha = 0.225$ ,  $\beta = 0.753$ ; daily s.d. 0.121%, kurtosis 27.8
- ▣ Interpretation: the half-life of a log-volatility shock is about 16 days
  - ▶ quiet periods with rare bursts are what a managed float produces
  - ▶ the high kurtosis says that Gaussian  $\epsilon_t$  is not enough (B4)

## B4: the particle filter for the EUR/RON SV model [Proposed]

- **Question:** is SV better than GARCH for EUR/RON on the likelihood scale, and is the particle likelihood reliable here?
- **Inputs:** the data and the posterior means of B3; bootstrap particle filter with systematic resampling; model: B3 and A7
- **Tasks**
  1. Run the filter with  $N = 500, 2000, 10\,000$  particles, 30 times each, and report the s.d. of  $\ln \hat{L}$ .
  2. Report the minimum and median ESS and the date of the minimum.
  3. Compare  $\ln \hat{L}$  (large  $N$ ) with the GARCH maximum.
  4. Interpretation: what does the ESS collapse tell you about the model?
- **Report:** three standard deviations, two ESS values and a date, two log-likelihoods, one sentence
- **Notebook:** section B4

## B5: the Romanian output gap in real time [Proposed]

- **Question:** how much does the end-of-sample estimate of the Romanian output gap change when later data arrive?
- **Inputs:** the data of B1; smooth-trend UC with AR(2) cycle (`fit_smooth_trend`); quasi-real time: re-estimate on data up to each quarter from 2012Q4 (parameters every 4 quarters) and keep the filtered gap; model: B1 and A10
- **Tasks**
  1. Compute the series of real-time (filtered, end-of-sample) gaps and the final smoothed gap.
  2. Compute the Orphanides–van Norden statistics: correlation, noise-to-signal ratio (s.d. of revisions over s.d. of the final gap), share of opposite signs.
  3. Report the largest revision.
  4. Interpretation: can monetary policy rely on the current estimate of the gap?
- **Report:** four statistics, one number, one sentence
- **Notebook:** section B5

## C1: is Romanian trend inflation anchored? [Proposed]

- ▣ **Question:** is the trend of Romanian inflation inside the BNR target band ( $2.5\% \pm 1$  pp), and how robust is the answer?
- ▣ **Inputs:** quarterly HICP inflation (% a.r.); UC-SV of Stock and Watson (2007) with quarterly seasonal dummies (ucsv\_gibbs); model: B3
- ▣ **Tasks**
  1. Write a five-line pre-registration: hypothesis, data, the anchoring statistic, the robustness grid, what would falsify the claim.
  2. Estimate the model for start dates 2001 and 2006 and for  $\gamma = 0.1$  and  $0.2$ ; report the trend in the last quarter with a 68% band.
  3. Compute the posterior probability that the trend lies in 1.5–3.5% in the last quarter and in 2019Q4.
  4. Interpretation: what can and cannot be concluded, and how would you turn this into a project?
- ▣ **Report:** a five-line plan, a table of four rows, a project plan
- ▣ **Notebook:** section C1

## C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise state space estimation. The answer:
- ▣ (a) The Kalman filter is useless if the errors are not Gaussian.
- ▣ (b) For a random-walk state, set  $P_1 = 10^{10}$  and keep all  $n$  terms of the likelihood.
- ▣ (c) An ML estimate of the level variance equal to zero proves that the level is constant.
- ▣ (d) SV and GARCH are the same model, since both make volatility persistent.
- ▣ (e) The particle-filter likelihood is unbiased, so its logarithm is unbiased too.
- ▣ (f) The Durbin-Koopman simulation smoother needs only one state-smoother pass per draw.
- ▣ (g) With an AR(1) cycle, the correlation between trend and cycle shocks can be estimated.
- ▣ Tasks
  - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, a number from today's tasks (notebook section C2).
  - ▶ 2. Report: a list of seven verdicts with one line of justification each.

## Key takeaways

- ▣ The exact diffuse filter lets the first observations fix the nonstationary states; they carry no information about the variances
- ▣ Sampling the states is the Kalman machinery run backwards (FFBS) or once more on simulated data (DK)
- ▣ SV adds a shock to the variance; particle filters estimate its likelihood, unbiased in levels only
- ▣ Trend growth, gaps and anchored trends are estimates with wide bands: report them, and report real-time values
- ▣ An AI answer is a draft: check the initialisation, the boundary, the log-likelihood bias and the identification

## After the seminar

- Lecture 6 develops each topic of today: the exact diffuse filter, the likelihood at the boundary, the three samplers, KSC, particle filters and PMMH, TVP, DFM, MNZ and UC-SV
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: trend inflation and anchoring in Central and Eastern Europe
- Reading: [Durbin and Koopman \(2012\)](#), Ch. 4–5; [Kim, Shephard and Chib \(1998\)](#); [Morley, Nelson and Zivot \(2003\)](#); [Stock and Watson \(2007\)](#)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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